

Precise INDUSTRIAL TIMER

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Industrial timers are widely used in the industry for automatic controlling of electrical devices. Most timers available in the market are either complicated or poor in quality with unsatisfactory cost-to-performance ratio. This DIY industrial timer is panel-mountable and offers excellent time precision for industrial use. The 7-segment display to show counting is directly driven by the microcontroller (MCU) without driver ICs.

Circuit and working

The circuit diagram of the industrial timer is shown in Fig. 1. It is built around 5V voltage regulator 7805 (IC1), AT89C52 MCU (IC2), 5V relay (RL1), two thumbwheel switches

(TWS1 and TWS2) and a few discrete components.

The circuit is powered by 230V AC mains. Transformer (X1) reduces 230V to 7.5V-0-7.5V. Diodes D1 and D2 are rectifier diodes, and capacitor C1 is connected as a filter. Filtered and regulated DC voltages are then fed to the circuit. AT89C52 MCU runs with 12MHz clock frequency, interfaced with two thumbwheel switches, two common-anode 7-segment displays (DIS1 and DIS2) and one 5V relay.

Transistor T1 works as a relay driver for RL1. Port pins P0 and P2 of IC2 are connected to 7-segment displays (P0 is MSB and P2 is LSB). LED2 is connected as a power-on indicator, and LED1 is connected on P3.0 pin as a timer-on indicator.

Switch S1 is used to turn on and off the power to the circuit. C4 and C5 are used for suppressing high-frequency signals generated by IC1. C6 and C7 are decoupling capacitors for the crystal. C2 and resistor R3 are power-on reset circuits for MCU IC2.

DIS1 and DIS2 are used to display time settings using TWS1 and TWS2. Details of TWS1 and TWS2 are shown in Fig. 2.

When S1 is closed, display indicates 00. After the timer starts counting up to set time, it automatically stops and the relay energises to cut off load. The display indicates the last count value till the timer restarts again.

Maximum count shown by DIS1 and DIS2 is 99 seconds. For example,

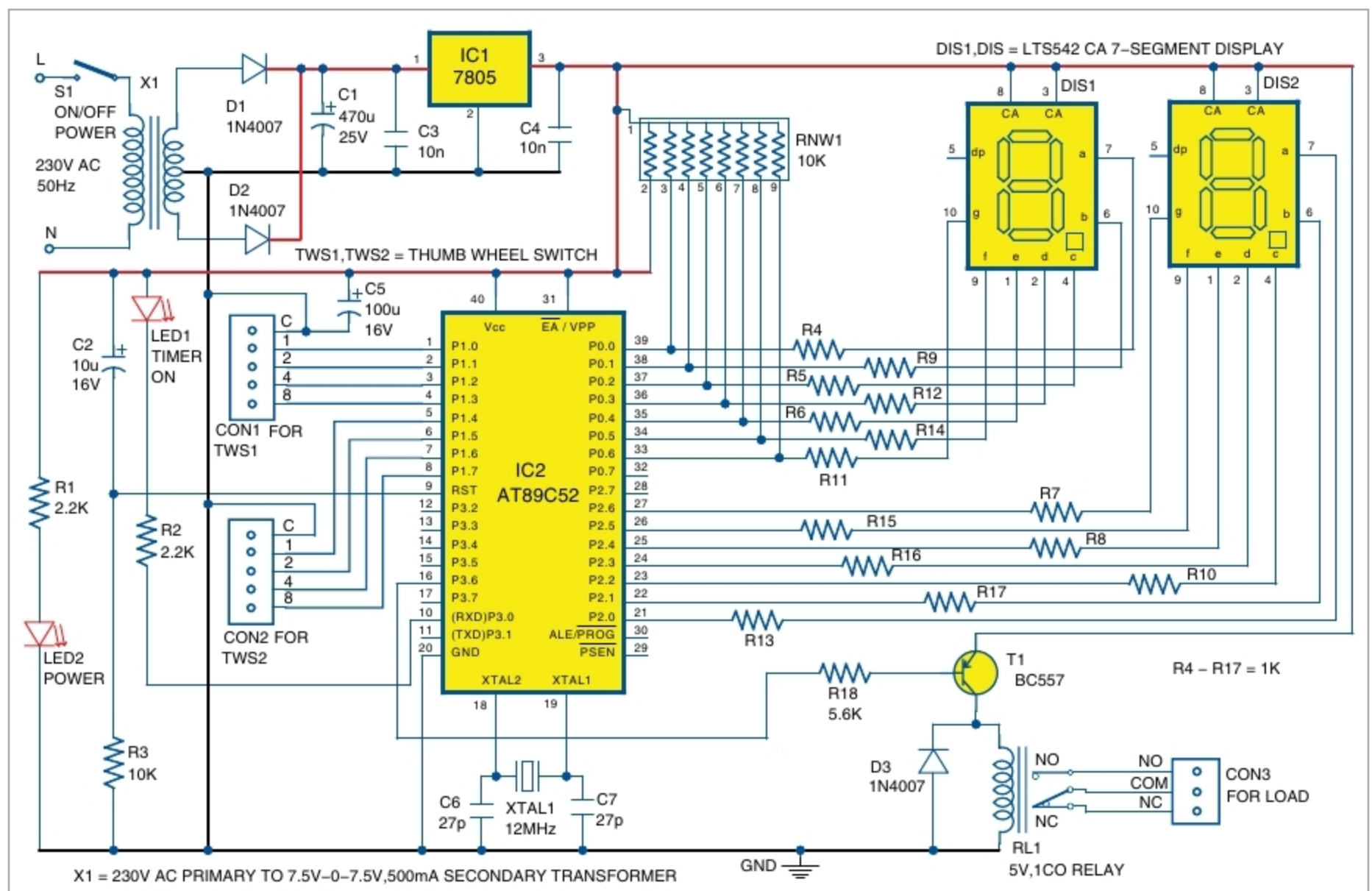


Fig. 1: Circuit diagram of industrial timer